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CHE 321; Spring - 2020; Quiz - 1

100 Points

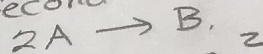
1. (25 points) What is the meaning of 'order of reaction'?

It is the biggest stoichiometric coefficient for the reactants and the concentration of that reactant gets multiplied exponentially to get the rate of reaction meaning at what rate that reactant gets consumed.

example: First order reaction $A \rightarrow B$

$$-r_A = kC_A^1$$

second order rxn.



$$-r_A = kC_A^2$$

2. (25 points) For a single liquid-phase reaction, which reactor will lead to a higher conversion: A plug-flow reactor (PFR) or a continuous stirred tank reactor (CSTRs)?

25 PFR

CSTR

3. (25 points) A catalytic CSTR reactor of volume
- V
- (
- m^3
-) has
- W
- (Mass) Kgs of the catalyst. What is the relationship between
- k_A
- and
- k'_A
- for this reactor? Note, the bulk density of the catalyst is
- $\rho_B = W/V$
- .

25
$$r_A = \frac{k_A}{k'_A} = \frac{\text{mass Kgs of catalyst (Kgs)}_A}{\text{volume (m}^3\text{)}} \cdot \frac{1}{k'_A}$$

4. (25 points) Qualitatively, draw a Levenspiel's plot for two CSTRs in series.

